

Answer: A

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Gravitational force = centripetal force

$$\frac{GM^2}{r^2} = \frac{Mv^2}{\left(\frac{r}{2}\right)}$$

$$v = \sqrt{\frac{GM}{2r}}$$

$$= \sqrt{\frac{(6.67 \times 10^{-11})(4.0 \times 10^{30})}{2(2.0 \times 10^{11})}}$$

$$= 2.6 \times 10^4 \text{ m s}^{-1}$$

Answer: B

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Max GPE = Total Energy = Max KE